

We consider the function

$$f(x, y) = \begin{cases} y^{-2} & \text{for } 0 < x < y \leq 1 \\ -x^{-2} & \text{for } 0 < y < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

First we note that each part of the function is continuous and thus by the tuborg rule the function is measurable

a) Calculate the 2 integrals

1)

$$\begin{aligned} \int_0^1 \left(\int_0^1 f(x, y) \lambda(dx) \right) \lambda(dy) \\ \int_0^1 f(x, y) \lambda(dx) &= \int_0^1 f(x, y) \mathbb{I}_{\{0 < x < y \leq 1\}} \lambda(dx) + \int_0^1 f(x, y) \mathbb{I}_{\{0 < y < x \leq 1\}} \lambda(dx) \\ &= \int_0^y y^{-2} \lambda(dx) + \int_y^1 -x^{-2} \lambda(dx) \end{aligned}$$

Since both functions are continuous on the bounded set we can simply calculate the Riemann integrals

$$\begin{aligned} &= R \int_0^y y^{-2} dx + R \int_y^1 -x^{-2} dx \\ &= \frac{1}{y} + 1 - \frac{1}{y} = 1 \\ \int_0^1 \left(\int_0^1 f(x, y) \lambda(dx) \right) \lambda(dy) &= \int_0^1 1 \lambda(dy) = 1 \cdot \lambda((0, 1]) = 1 \end{aligned}$$

2)

$$\int_0^1 \left(\int_0^1 f(x, y) \lambda(dy) \right) \lambda(dx)$$

We calculate the "inner" integral

$$\begin{aligned} \int_0^1 f(x, y) \lambda(dy) &= \int_0^1 f(x, y) \mathbb{I}_{\{0 < x < y \leq 1\}} \lambda(dy) + \int_0^1 f(x, y) \mathbb{I}_{\{0 < y < x \leq 1\}} \lambda(dy) \\ &= \int_x^1 y^{-2} \lambda(dy) + \int_0^x -x^{-2} \lambda(dy) \end{aligned}$$

Again since we have continuous functions we can simply calculate the Riemann integrals

$$\begin{aligned} &= R \int_x^1 y^{-2} dy + R \int_0^x -x^{-2} dy \\ &= -1 + \frac{1}{x} - \frac{1}{x} = -1 \end{aligned}$$

$$\int_0^1 \left(\int_0^1 f(x, y) \lambda(dy) \right) \lambda(dx) = \int_0^1 -1 \cdot \lambda(dx) = -1 \cdot \lambda((0, 1]) = -1$$

b) Determine if $f \in \mathcal{L}^1(\lambda_2)$

No we cannot have $f \in \mathcal{L}^1(\lambda_2)$

Since Fubini's theorem states that

$$f \in \mathcal{L}^1(\lambda_2) \Rightarrow \int_0^1 \left(\int_0^1 f(x, y) \lambda(dy) \right) \lambda(dx) = \int_0^1 \left(\int_0^1 f(x, y) \lambda(dx) \right) \lambda(dy)$$

Since $1 \neq -1$

We must have $f \notin \mathcal{L}^1(\lambda_2)$